

Take Home Activity 07

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Q1

```
public class Calculator {

    int add(int a, int b) {
        int result = a + b;
        System.out.println("Int sum: " + result);
        return result;
    }

    double add(double a, double b) {
        double result = a + b;
        System.out.println("Double sum: " + result);
        return result;
    }

    int add(int a, int b, int c) {
        int result = a + b + c;
        System.out.println("Three-int sum: " + result);
        return result;
    }

    String add(String a, String b) {
        String result = a + b;
        System.out.println("String join: " + result);
        return result;
    }
}
```

```
public class Main {
    public static void main(String[] args) {

        Calculator calc = new Calculator();

        calc.add(10, 20);
        calc.add(5.5, 4.5);
        calc.add(10, 20, 30);
    }
}
```

```
        calc.add("Hello ", "Java");  
    }  
}
```

```
Int sum: 30  
Double sum: 10.0  
Three-int sum: 60  
String join: Hello Java
```

Q2

```
package Q2;  
  
public class Printer {  
  
    void printData(String text) {  
        System.out.println("Printing text: " + text);  
    }  
  
    void printData(int number) {  
        System.out.println("Printing number: " + number);  
    }  
  
    void printData(String text, int copies) {  
        System.out.println("Printing " + copies + "copies of: " +  
text);  
    }  
  
    void printData(boolean isDuplex) {  
        System.out.println("Duplex printing: " + isDuplex);  
    }  
}
```

```

package Q2;

public class Main {
    public static void main(String[] args) {
        Printer p1 = new Printer();

        p1.printData("Hello");
        p1.printData(18);
        p1.printData("Navitha,18");
        p1.printData(true);
    }
}

```

```

a125b018f6704d209e6ec7\redhat.3
Printing text: Hello
Printing number: 18
Printing text: Navitha,18
Duplex printing: true

```

Q3

```

package Q3;

public class Animal {
    String name;
    void makeSound(){
        System.out.println(name + "makes a generic sound");
    }
}

class Dog extends Animal {
    @Override
    void makeSound(){
        System.out.println(name + " says Woof! ");
    }
}

```

```
class Cat extends Animal {
    @Override
    void makeSound() {
        System.out.println(name + " says Meow! ");
    }
}

class Bird extends Animal {
    @Override
    void makeSound() {
        System.out.println(name + " says Tweet! ");
    }
}
```

```
package Q3;

public class Main {
    public static void main(String[] args) {
        Dog dog = new Dog();
        dog.name = "Rex";
        dog.makeSound();

        Cat cat = new Cat();
        cat.name = "Luna";
        cat.makeSound();

        Bird bird = new Bird();
        bird.name = "Tweety";
        bird.makeSound();

        Animal a1 = new Dog();
        a1.name = "Bruno";
        a1.makeSound();

        Animal a2 = new Cat();
        a2.name = "Kitty";
        a2.makeSound();

        Animal a3 = new Bird();
        a3.name = "Sky";
        a3.makeSound();
    }
}
```

```
}  
}
```

```
a1230018107040209c0cc7 \r  
Rex says Woof!  
Luna says Meow!  
Tweety says Tweet!  
Bruno says Woof!  
Kitty says Meow!  
Sky says Tweet!
```

Q4

```
package Q4;  
  
public class Shape {  
    void draw() {  
        System.out.println("Drawing a generic shape");  
    }  
  
    void area() {  
        System.out.println("Area: 0.0");  
    }  
}
```

```
package Q4;  
  
public class Circle extends Shape {  
    double radius;  
  
    Circle(double radius) {  
        this.radius = radius;  
    }  
}
```

```

@Override
void draw() {
    System.out.println("Drawing a Circle");
}

@Override
void area() {
    double result = Math.PI * radius * radius;
    System.out.println("Area: " + result);
}
}

```

```

package Q4;

public class Rectangle extends Shape {
    double width;
    double height;

    Rectangle(double width, double height) {
        this.width = width;
        this.height = height;
    }

    @Override
    void draw() {
        System.out.println("Drawing a Rectangle");
    }

    @Override
    void area() {
        double area = width * height;
        System.out.println("Area: " + area);
    }
}

```

```

package Q4;

public class Triangle extends Shape {
    double base;

```

```

double height;

Triangle(double base, double height) {
    this.base = base;
    this.height = height;
}

@Override
void draw() {
    System.out.println("Drawing a Triangle");
}

@Override
void area() {
    double area = 0.5 * base * height;
    System.out.println("Area: " + area);
}
}

```

```

package Q4;

public class Main {
    public static void main(String[] args) {
        Shape[] shapes = new Shape[3];

        shapes[0] = new Circle(5);
        shapes[1] = new Rectangle(4, 6);
        shapes[2] = new Triangle(3, 8);

        for (Shape s : shapes) {
            s.draw();
            s.area();
            System.out.println();
        }
    }
}

```

Drawing a Circle

Area: 78.53981633974483

Drawing a Rectangle

Area: 24.0

Drawing a Triangle

Area: 12.0

Q5

```
package Q5;

public class BankAccount {
    String accountNumber;
    String holderName;
    double balance;

    void showDetails(){
        System.out.println("Account: " +accountNumber+ " Holder: "
+holderName+ " Balance " +balance);
    }

    void deposit( double amount){
        balance = balance + amount;
        System.out.println("New balance: " + balance);
    }

    void deposit(double amount, String currency){
        if (currency == "USD"){
            balance += (amount * 300);
            System.out.println("Deposited " + amount + currency + " = "
+ " LKR " + amount * 300 + ". New Balance: " + balance);
        }
    }
}
```

```
}
```

```
package Q5;

public class SavingsAccount extends BankAccount {
    double interestRate;
    @Override
    void showDetails(){
        System.out.println("Account: " + accountNumber + " | Holder: "
+ holderName + " | Balance: " + balance + " | Type: Savings | Rate: " +
interestRate + "%");
    }
    void applyInterest(){
        balance += balance * interestRate / 100;
        System.out.println("Interest applied. New Balance: " + balance
);
    }
}
```

```
package Q5;

public class LoanAccount extends BankAccount{
    double loanAmount;

    @Override
    void showDetails(){
        System.out.println("Account: " + accountNumber + " | Holder: " +
holderName + " | Balance: " + balance + " | Type: Loan | Loan: " +
loanAmount);
    }

    void repayLoan(double amount){
        loanAmount -= amount;
        System.out.println("Loan repaid: " + amount + ". Remaining: " +
loanAmount);
    }
}
```

```

package Q5;

public class Main {
    public static void main(String[] args) {

        SavingsAccount s1 = new SavingsAccount();
        s1.accountNumber = "SA-001";
        s1.holderName = "Alice";
        s1.balance = 5000;
        s1.interestRate = 3.5;

        s1.showDetails();
        s1.deposit(1000.0);
        s1.deposit(50.0, "USD");
        s1.applyInterest();

        System.out.println();

        BankAccount b1 = new SavingsAccount();
        b1.accountNumber = "SA-001";
        b1.holderName = "Alice";
        b1.balance = 5000;

        ((SavingsAccount) b1).interestRate = 3.5;
        b1.showDetails();

        BankAccount b2 = new LoanAccount();
        b2.accountNumber = "LA-002";
        b2.holderName = "Bob";
        b2.balance = 0;

        ((LoanAccount) b2).loanAmount = 20000;
        b2.showDetails();
    }
}

```

```

Account: SA-001 | Holder: Alice | Balance: 5000.0 | Type: Savings | Rate: 3.5%
New balance: 6000.0
Deposited 50.0USD = LKR 15000.0. New Balance: 21000.0
Interest applied. New Balance: 21735.0

```

```

Account: SA-001 | Holder: Alice | Balance: 5000.0 | Type: Savings | Rate: 3.5%
Account: LA-002 | Holder: Bob | Balance: 0.0 | Type: Loan | Loan: 20000.0

```

Q6

```
package Q6;

public class SecurityAlert {
    String alertId;
    String source;
    String severity;

    void logAlert() {
        System.out.println("[ALERT] ID: " + alertId + " | Source: " +
source + " | Severity: " + severity);
    }

    void logAlert(String extraInfo) {
        System.out.println("[ALERT] ID: " + alertId + " | Source: " +
source + " | Severity: " + severity + " | Info: " + extraInfo);
    }

    void logAlert(String extraInfo, boolean sendEmail) {
        System.out.println("[ALERT] ID: " + alertId + " | Source: " +
source + " | Severity: " + severity + " | Info: " + extraInfo);

        if (sendEmail) {
            System.out.println("Email notification sent for " +
alertId);
        }
    }

    void raiseAlert() {
        System.out.println("!! SECURITY ALERT !! " + alertId + " from "
+ source);
    }
}
```

```
package Q6;

public class MalwareAlert extends SecurityAlert {
    String malwareType;

    @Override
```

```

    void raiseAlert() {
        System.out.println("[MALWARE] " + malwareType + " detected from " + source + "!");
    }

    void quarantine() {
        System.out.println("Quarantining " + malwareType + " from " + source + "...");
    }
}

```

```

package Q6;

public class PhishingAlert extends SecurityAlert {
    String targetEmail;

    @Override
    void raiseAlert() {
        System.out.println("[PHISHING] Attack targeting " + targetEmail + " from " + source + "!");
    }

    void blockEmail() {
        System.out.println("Blocking phishing email targeting: " + targetEmail);
    }
}

```

```

package Q6;

public class DDoSAlert extends SecurityAlert{
    int packetCount;

    @Override
    void raiseAlert() {
        System.out.println("[DDoS] " + source + " - " + packetCount + " packets!");
    }
}

```

```

    void mitigate() {
        System.out.println("Mitigating DDoS from " + source + " -" +
packetCount + " packets dropped");
    }
}

```

```

package Q6;

public class Main {
    public static void main(String[] args) {

        MalwareAlert m1 = new MalwareAlert();
        m1.alertId = "ALT-001";
        m1.source = "192.168.1.5";
        m1.severity = "Critical";
        m1.malwareType = "Ransomware";

        m1.logAlert();
        m1.logAlert("Encrypted 3 files");
        m1.logAlert("Encrypted 3 files", true);
        m1.raiseAlert();
        m1.quarantine();

        System.out.println();

        SecurityAlert[] alerts = new SecurityAlert[3];

        MalwareAlert malware = new MalwareAlert();
        malware.alertId = "ALT-001";
        malware.source = "192.168.1.5";
        malware.severity = "Critical";
        malware.malwareType = "Ransomware";

        PhishingAlert phishing = new PhishingAlert();
        phishing.alertId = "ALT-002";
        phishing.source = "evil.com";
        phishing.severity = "High";
        phishing.targetEmail = "admin@nsbm.ac.lk";

        DDoSAlert ddos = new DDoSAlert();
        ddos.alertId = "ALT-003";
    }
}

```

```
        ddos.source = "10.0.0.99";
        ddos.severity = "Critical";
        ddos.packetCount = 50000;

        alerts[0] = malware;
        alerts[1] = phishing;
        alerts[2] = ddos;

        for (SecurityAlert alert : alerts) {
            alert.logAlert();
            alert.raiseAlert();
            System.out.println();
        }
    }
}
```

```
a125b018f6704d209e6ec7\redhat.java\jdt_ws\Java-Labs-NSBM---25.3_f9ac9813\bin' 'Q6.Main'
[ALERT] ID: ALT-001 | Source: 192.168.1.5 | Severity: Critical | Info: Encrypted 3 files
Email notification sent for ALT-001
[MALWARE] Ransomware detected from 192.168.1.5!
Quarantining Ransomware from 192.168.1.5...

[ALERT] ID: ALT-001 | Source: 192.168.1.5 | Severity: Critical
[MALWARE] Ransomware detected from 192.168.1.5!

[ALERT] ID: ALT-002 | Source: evil.com | Severity: High
[PHISHING] Attack targeting admin@nsbm.ac.lk from evil.com!

[ALERT] ID: ALT-003 | Source: 10.0.0.99 | Severity: Critical
[DDoS] 10.0.0.99 - 50000 packets!
```